

교육용 자료 안내

본 자료는 Darun Color Institute의 색채 교육 과정을 위해 제공되는 교육용 문서입니다. 학생들이 안전한 재료 취급을 학습하기 위해 사용된 원료의 물리화학적 특성과 안전 정보를 공개합니다.

다만, 이 자료에 포함된 정보는 교육 목적으로만 활용되어야 하며, 상업적 목적의 제품 제조 및 판매는 엄격히 금지됩니다.

화장품 제조업 허가를 취득한 자가 독립적으로 제품을 개발·판매할 경우, 다음을 반드시 준수해야 합니다:

- 신뢰할 수 있는 화장품 원료 전문 제조사로부터 원료를 직접 확보
- 해당 제조사로부터 최신 MSDS 및 시험성적서(CoA, 안정성 시험 등)를 개별 요청
- 식품의약품안전처 규정에 따른 원료 검증 및 제품 신고

이는 소비자 안전과 법적 책임을 보장하기 위한 필수 절차입니다.

Educational Material Notice

This document is provided by Darun Color Institute as an educational resource for our color science training programs. To support student learning in safe material handling practices, we disclose the physical, chemical, and safety characteristics of the raw materials used in our formulations.

However, all information contained in this material is strictly for educational purposes only. Any commercial use for product manufacturing or sales is strictly prohibited.

Should you obtain cosmetic manufacturing authorization and independently develop products for commercial purposes, you must:

- Source raw materials directly from reliable cosmetic ingredient manufacturers
- Request current MSDS and certificates of analysis (CoA) individually from the respective manufacturers
- Verify all raw materials and complete product registration in compliance with Korea Food and Drug Administration (KFDA) regulations

These procedures are mandatory requirements to ensure consumer safety and fulfill legal responsibilities.

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Material Safety Data Sheet (MSDS)

1. Chemical Product and Company Identification

Item	Information
Product Name	Educational Z-Brown Powder
Recommended Use	Cosmetic raw material
Means of Identification	Not available
Relevant identified uses	Not applicable

2. Hazards Identification

Category	Details
GHS Classification	Specific Target Organ Toxicity (Single Exposure) - Category 3 (Respiratory irritation)
Hazard Statements	H335: May cause respiratory irritation
Precautionary Statements	P261: Avoid inhalation of dust, fumes, gas, mist, vapor, spray P271: Use only outdoors or in well-ventilated areas P304+P340: If inhaled: Move to fresh air and rest in comfortable position P312: Seek medical advice if you feel unwell P403+P233: Store in well-ventilated place. Keep container tightly closed P405: Store in locked storage P501: Dispose of contents/container according to regulations

3. Composition/Information on Ingredients

**Composition ratios are provided as approximate ranges for educational purposes*

Substance/mixture : Mixture

There are no ingredients present which, within the current knowledge of the supplier and in the concentrations applicable, are classified as hazardous to health or the environment, are PBTs or vPvBs or have been assigned a workplace exposure limit and hence require reporting in this section.

4. First Aid Measures

Route of Exposure	First Aid Procedure
Eye Contact	Seek emergency medical attention. Rinse thoroughly with water for at least 20 minutes. If symptoms develop, seek medical examination.
Skin Contact	Seek medical attention if skin disorder symptoms persist. Remove contaminated clothing and shoes before reuse. Rinse with running water for at least 20 minutes if needed.
Inhalation	If excessive dust or fumes exposure occurs, remove to clean air. Take coughing or other abnormal symptoms seriously and seek medical

	examination.
Ingestion	Do not induce vomiting. For ingestion, seek immediate medical attention.
Medical Attention	Ensure medical personnel are aware of substance exposure and take appropriate protective measures.

5. Fire-Fighting Measures

Item	Information
Suitable Extinguishing Media	Alcohol foam, carbon dioxide, water spray, dry sand for small fires
Large Fire Method	Move container away from fire area if safe. Cool containers with large amounts of water even after extinguishing.
Specific Hazards	Container may explode when heated. Some materials can burn but do not ignite easily. Irritating, corrosive, and toxic gases may be generated during combustion.
Protective Equipment	Wear self-contained breathing apparatus and chemical-resistant protective clothing.

6. Accidental Release Measures

Action	Description
Prevention	Prevent dust formation and scattering. Use full or local exhaust ventilation.
Environmental Protection	Prevent entry into waterways, sewers, basements, and confined spaces. Spillage may cause contamination.
Cleanup Methods	For small spills: Absorb with inert material and place in container. For large spills: Create barriers to prevent spreading. Cover powder spills with plastic sheet to prevent diffusion and keep dry.

7. Handling and Storage

Category	Requirements
Handling	Avoid dust inhalation. Handle with care. Use full or local exhaust ventilation. Be cautious when opening containers. Follow all protective measures after handling.
Storage	Store in tightly closed container. Store in cool, dry, and well-ventilated place. Store away from incompatible materials. Empty drums should be completely drained and properly sealed before returning.

8. Exposure Controls/Personal Protection

Control Type	Requirement
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Exposure Limits	Iron Oxide (Fe ₂ O ₃): TWA 5 mg/m ³
Engineering Controls	Maintain ventilation during operation to keep air pollution below exposure limits. Ensure work processes meet labor standards.
Respiratory Protection	Wear KOSHA-certified respiratory protection appropriate for exposed particulate matter. For concentrations below 50 mg/m ³ , use appropriate filter-equipped half-mask respirator.
Eye Protection	Use safety goggles. Install emergency eyewash and shower facilities near work area.
Hand Protection	Wear appropriate chemical-resistant protective gloves to prevent direct contact.
Body Protection	Wear appropriate chemical-resistant protective clothing to prevent skin exposure.

9. Physical and Chemical Properties

Property	Value
Appearance	Solid (powder) - Brown to black color
Odor	Odorless
pH	No data available
Melting Point	1565°C (Iron Oxide)
Solubility	Insoluble in water; soluble in acid
Specific Gravity	Iron Oxide: 5.24, Iron Hydroxide Oxide: 0.97 (estimated)

10. Stability and Reactivity

Item	Information
Chemical Stability	Stable at room temperature and normal pressure
Conditions to Avoid	Heat, sparks, flames, and other ignition sources
Incompatible Materials	Oxidizing agents
Hazardous Decomposition	Container may explode when heated. Materials are non-flammable but can burn under certain conditions. Irritating, corrosive, and toxic gases may be generated during combustion.

11. Toxicological Information

Effect	Description
Acute Toxicity (Oral)	LD50 > 5000 mg/kg (Rat)
Acute Toxicity (Inhalation)	Dust LC50: 5.05 mg/L 4hr (Rat)
Skin Irritation	OECD 404: No irritation observed
Eye Irritation	OECD 405: No irritation observed
Respiratory Sensitization	No data available

Skin Sensitization	Guinea pig sensitization: Non-sensitizing
Carcinogenicity	Iron Oxide: Group A4 (ACGIH)
Respiratory Irritation (Single Exposure)	May cause respiratory irritation
Target Organ Toxicity (Repeated Exposure)	Lung weight increase observed with prolonged exposure. NOAEC ≥ 30 mg/m ³
Aspiration Hazard	No data available

12. Ecological Information

Category	Information
Fish Toxicity	LC96hr ≥ 50000 mg/L (Danio rerio)
Crustacean Toxicity	EC50 > 100 mg/L 48hr (Daphnia magna)
Bird Toxicity	No data available
Persistency	No data available
Degradability	No data available
Bio-condensability	No data available
Soil Mobility	No data available
Other Effects	No data available

13. Disposal Considerations

Item	Information
Disposal Method	1) Treat by neutralization, hydrolysis, oxidation, or reduction 2) Incinerate at high temperature or treat by high-temperature fusion 3) Solidify and dispose
Precautions	Dispose of contents and container according to Waste Management Act regulations

14. Transport Information

Item	Information
UN Number	Iron Oxide (spent): UN 1376
Proper Shipping Name	No UN hazardous materials classification
Transport Hazard Class	Not applicable
Packing Group	Not applicable
Marine Pollutant	Not applicable

15. Regulatory Information

Item	Information
Regulatory Basis	Prepared in accordance with Korean Occupational Safety and Health Act and relevant regulations. Iron Oxide is a monitored substance (measurement period: dust and fumes) and special health check subject (12-month cycle).

16. Other Information

Item	Information
Disclaimer	The information in this document is based on accurate data and prepared with reference to MSDS from Korea Occupational Safety and Health Agency. However, we do not guarantee the accuracy of results or values obtained when using this material for specific purposes or mixing with other substances. The company assumes no responsibility for personal losses or property damage to buyers or third parties.
Prepared by	Darun Color Institute
Document Type	For Educational Purposes Only